

1.2 Functions

Pure Mathematics 1 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A function assigns each allowed input one output. State domain before finding range or inverse; inverse exists on a one-to-one restricted domain. A composite fg means $f(g(x))$ under common notation, with domain restricted so $g(x)$ lies in f 's domain.

Worked example: follow the reasoning

For $f(x)=2x+1$ and $g(x)=x^2$, $fg(x)=2x^2+1$ whereas $gf(x)=(2x+1)^2$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 For $f(x)=3x-2$, find $f^{-1}(x)$ and $f^{-1}(10)$.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

$y=3x-2$ gives $x=(y+2)/3$ [2]; $f^{-1}(10)=4$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.